

A randomized trace benchmark has variance $1/m$

Conjecture 00000008800 ziangni-sys

The source claims that seed variance in reproducible randomized numerical benchmarks is the reciprocal square root of the repetition count. The distinction between variance and standard deviation matters: an ordinary nondegenerate randomized trace estimator has variance $1/m$, not $1/\sqrt{m}$, after averaging m independent repetitions. The discrepancy persists at the level of asymptotic order.

Counterexample. *There is an unbiased randomized matrix-trace benchmark with independent uniform seeds for which the averaged estimator $\hat{\tau}_m$ satisfies*

$$\text{Var}(\hat{\tau}_m) = \frac{1}{m}, \quad \frac{\text{Var}(\hat{\tau}_m)}{1/\sqrt{m}} \rightarrow 0.$$

In particular, its variance admits no eventually positive constant multiple of $1/\sqrt{m}$ as a lower bound.

Proof. Fix the actual matrix and diagonal-sampling algorithm

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad J \text{ uniform on } \{0, 1\}, \quad X(J) = 2A_{JJ}.$$

Thus X equals 2 or 0, each with probability $1/2$. It is reproducible from its seed J , and

$$\mathbb{E}X = 1 = \text{tr } A, \quad \text{Var}(X) = \frac{1}{2}(2 - 1)^2 + \frac{1}{2}(0 - 1)^2 = 1.$$

For each integer $m \geq 1$, use the finite product probability space

$$\Omega_m = \{0, 1\}^{\{1, \dots, m\}}, \quad \mu_m = \left(\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1\right)^{\otimes m}.$$

The coordinate seeds $J_1, \dots, J_m$ are independent and uniformly distributed. Define $X_i = X(J_i)$ and the reported benchmark estimate

$$\hat{\tau}_m = \frac{1}{m} \sum_{i=1}^m X_i.$$

Its expectation is again $\text{tr } A = 1$. Independence and the actual one-run variance give

$$\text{Var}(\hat{\tau}_m) = \frac{1}{m^2} \sum_{i=1}^m \text{Var}(X_i) = \frac{1}{m}.$$

Therefore its ratio to the claimed reciprocal-root scale is $1/\sqrt{m}$, which tends to zero. For any $c > 0$, sufficiently large m satisfy

$$\text{Var}(\hat{\tau}_m) < \frac{c}{\sqrt{m}}.$$

This excludes both equality with the stated expression and a two-sided asymptotic law of that order. $\square$

The standard deviation in this example is exactly $1/\sqrt{m}$, consistent with the usual Monte Carlo scaling. The source explicitly uses variance in both languages; replacing it by standard deviation changes the claim. The counterexample uses the variance across independent seeds of the benchmark's arithmetic-mean output, a standard repetition-based benchmark quantity. It does not posit any failure of Bernoulli concentration inequalities.

Formal verification. Lean constructs the two-point probability mass function, the actual matrix and estimator, and every finite product measure. It proves uniform coordinate marginals, `iIndepFun` for the seeds and outputs, identical distribution with the one-run estimator, unbiasedness, and the exact `ProbabilityTheory.variance` for every positive repetition count. Finally it proves the ratio limit and explicitly negates every eventual positive reciprocal-root lower bound. The pinned Lean 4.19.0 project prints axiom audits of its main theorems.